* Defn A set A is called a proper subset of set B if $A \subset B$ and $A \neq B$. We write $A \subsetneq B$

* Defn $\mathbb{N}$: The set of natural numbers
$$\mathbb{N} := \{1, 2, 3, \ldots\}$$

* Defn Set theoretic difference of A and B / The complement of B relative to A.
$$A \setminus B := \{x : x \in A \text{ and } x \notin B\}$$

* Defn Sets A and B are called disjoint if
$$A \cap B = \emptyset$$

* Lemma
$$A \setminus (B \cup C) = (A \setminus B) \cap (A \setminus C)$$
$$A \setminus (B \cap C) = (A \setminus B) \cup (A \setminus C)$$

* Theorem (Principle of Induction). Let $P(n)$ be a statement depending on the natural number n. Suppose that
1. $P(1)$ is true
2. If $P(n)$ is true then $P(n+1)$ is true

Then $P(n)$ is true for all $n \in \mathbb{N}$

Hint: Use well ordering property of $\mathbb{N}$

* Defn Well ordering property of $\mathbb{N}$. Every non empty set of $\mathbb{N}$ has a smallest element.

* Defn Let A and B be sets. The Cartesian product is the set of tuples defined by
$$A \times B := \{(x, y) : x \in A, y \in B\}$$

* Defn A function $f : \overbrace{A}^{\text{domain of } f} \longrightarrow \overbrace{B}^{\text{codomain of } f}$ is the subset f of $A \times B$ s.t for each $x \in A$ $\exists$ a unique $y \in B$ for which $(x,y) \in f$. We write $f(x) = y$

$$\underbrace{R(f)}_{\text{range of } f} := \{ y \in B : \exists x \in A \text{ s.t } f(x) = y \}$$

* Defn $f : A \longrightarrow B$. Image / Direct Image of $C \subset A$

$$f(C) := \{ f(x) \in B : x \in C \}$$

Inverse Image of $D \subset B$

$$f^{-1}(D) := \{ x \in A : f(x) \in D \}$$

* Lemma $f : A \longrightarrow B$. Let C, D be subsets of B then